

Generalizing Remaining Useful Life Prediction of TURBOFAN Jet Engine via Domain Adaptation Techniques

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Abstract—Accurate prediction of remaining useful life (RUL) in aircraft engines is crucial for effective predictive maintenance and operational efficiency. Traditional machine learning models often struggle to generalize across different engine operating conditions and fault modes, requiring extensive retraining for each new dataset. This project explores domain adaptation techniques to improve the generalization of RUL prediction models across multiple datasets in the NASA C-MAPSS dataset. Using methods such as Batch Normalization (BN) and Domain-Adversarial Neural Networks (DANN), I aim to align feature distributions and enhance model robustness across varying operating conditions. The proposed approach is expected to reduce computational costs while improving prediction accuracy, making it highly relevant for real-world aerospace applications. The effectiveness of domain adaptation will be evaluated using RMSE, MAE, and cross-domain transferability metrics such as Domain Classification Accuracy (DCA).

Index Terms—Domain adaptation, Remaining Useful Life (RUL), predictive maintenance, domain adversarial neural networks (DANN), batch normalization, NASA C-MAPSS dataset.

I. INTRODUCTION

Predictive maintenance has become a crucial area of research in the aerospace industry, where Remaining Useful Life (RUL) prediction plays a vital role in preventing unexpected failures and optimizing maintenance schedules. The NASA C-MAPSS dataset is widely used for developing and testing machine learning models for RUL estimation, as it provides extensive sensor readings under different engine conditions and fault scenarios [1]. However, one of the significant challenges in this domain is the lack of generalization across different datasets due to variations in operating conditions and fault modes. Traditional machine learning models trained on a single dataset often fail to perform well when applied to datasets with different conditions, necessitating extensive retraining and increasing computational costs.

II. STATE OF RESEARCH

Current research in the field of RUL prediction has focused on deep learning-based methods, including Long Short-Term

Memory (LSTM) networks and Convolutional Neural Networks (CNNs) [2]. While these models demonstrate high accuracy on individual datasets, they struggle with domain shifts, making them ineffective for cross-domain generalization. Recently, domain adaptation techniques have emerged as a promising solution to mitigate this issue. Methods such as Batch Normalization (BN) and Domain-Adversarial Neural Networks (DANN) help in aligning feature distributions across different domains, enabling models to generalize more effectively [3].

III. AIM

This project aims to develop a domain adaptation-based RUL prediction model that improves model robustness and adaptability across multiple datasets. By leveraging BN and DANN, the proposed approach seeks to align sensor data distributions from different operating conditions, reducing the reliance on dataset-specific retraining and enhancing the model's overall predictive performance. The research question guiding this project is: "How effectively can domain adaptation techniques enhance the generalization of RUL prediction models across varying operating conditions and fault modes in aircraft engines, minimizing the need for dataset-specific retraining?" The proposed method will be evaluated by comparing its performance on different datasets using standard RUL metrics and cross-domain transferability metrics.

IV. OBJECTIVES

- Train a baseline RUL prediction model (LSTM) on FD001.
- Evaluate its performance on FD002–FD004 without adaptation to measure degradation in accuracy.
- Implement Domain Adaptation techniques (Batch Normalization, DANN) to improve generalization.
- Compare baseline vs. adapted models in terms of accuracy, robustness, and computational efficiency.

V. EXPECTED OUTCOMES

- A model capable of cross-domain RUL prediction without requiring extensive retraining, enabling broader applicability in predictive maintenance for aircraft engines.

- Reduced computational costs for predictive maintenance applications, minimizing the need for repeated dataset-specific model training.
- Improved generalization across different jet engine types, allowing for more accurate and reliable predictions in various operational conditions.
- Enhanced operational efficiency and safety in the aviation industry by facilitating proactive maintenance scheduling and reducing unexpected failures.

VI. METHODOLOGY

To answer the research question, I will employ a combination of quantitative experiments, domain adaptation techniques, and comparative model evaluations. The methodology consists of three key stages:

A. Data Collection and Preprocessing

- Utilize the NASA C-MAPSS dataset (FD001–FD004) for training and testing.
- Normalize sensor readings and compute Remaining Useful Life (RUL) labels.
- Apply feature engineering to extract relevant sensor trends that contribute to RUL estimation.

B. Model Development and Domain Adaptation

- Train a baseline RUL prediction model (LSTM) on FD001.
- Evaluate its performance on FD002–FD004 without adaptation to assess domain shift effects.
- Implement Batch Normalization (BN) to align feature distributions across datasets.
- Apply Domain-Adversarial Neural Networks (DANN) to learn domain-invariant features and enhance model

transferability.

C. Performance Evaluation

The effectiveness of domain adaptation will be assessed using:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Domain Classification Accuracy (DCA)
- Maximum Mean Discrepancy (MMD)

VII. RISKS AND SOLUTIONS

The implementation of domain adaptation techniques for RUL prediction comes with several challenges. One significant risk is poor generalization across datasets, which can be mitigated by incorporating DANN and Batch Normalization to improve feature alignment. Another challenge is the computational complexity of deep learning models, which can be addressed by utilizing lightweight architecture such as Quantized LSTMs to balance efficiency and accuracy. Additionally, insufficient labeled failure data in some operating conditions can be overcome through data augmentation and semi-supervised learning techniques. Lastly, overfitting due to limited adaptation samples can be mitigated by employing regularization methods, dropout layers, and early stopping strategies.

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